

Comparative Evaluation of Recursive Partitioning and Causal Machine Learning Methods for Subgroup-Specific Treatment Effect Estimation

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ABSTRACT:

Heterogeneous treatment effects (HTE) are central to precision medicine, as they capture variations in treatment responses across patient subgroups. Accurate identification of these subgroups and reliable estimation of subgroup-specific treatment effects are critical for optimizing clinical decision-making. This study conducts a comparative evaluation of recursive partitioning methods (model-based trees), causal machine learning approaches (causal forests), and Bayesian models for subgroup-level treatment effect estimation. Using both simulated and real-world high-dimensional clinical datasets, we assess the methods in terms of subgroup identification accuracy, bias and variance of treatment effect estimates, interpretability, and computational efficiency. Results highlight the relative strengths and limitations of each approach, providing practical guidance for researchers and clinicians seeking to implement HTE analyses. This study contributes to methodological advances in precision medicine by offering a comprehensive comparison of modern approaches for subgroup-specific treatment effect estimation.

Keywords:

Heterogeneous treatment effects, Recursive partitioning, Model-based trees, Causal forests, Bayesian models, Subgroup analysis, Precision medicine, Treatment effect estimation, High-dimensional clinical data

1. INTRODUCTION

1.1 Background

In clinical research, treatments often do not have uniform effects across all patients. Differences in genetics, demographics, comorbidities, and other patient-specific characteristics can lead

to **heterogeneous treatment effects (HTE)**. Recognizing and quantifying this heterogeneity is fundamental to the implementation of **precision medicine**, where interventions are tailored to the unique profiles of patient subgroups to maximize therapeutic efficacy and minimize adverse effects.

Traditional statistical approaches, such as standard regression models, assume homogeneous treatment effects across populations and may fail to capture complex interactions between covariates and treatment responses. To address this limitation, modern approaches such as **recursive partitioning**, **causal machine learning**, and **Bayesian methods** have emerged, enabling more flexible modeling of HTE in both clinical trial and observational data.

1.2 Motivation

Despite the proliferation of these advanced methods, there is limited guidance on their **comparative performance** in identifying clinically meaningful subgroups and accurately estimating subgroup-specific treatment effects, particularly in **high-dimensional datasets** rich in biomarkers and covariates. Recursive partitioning methods, such as model-based trees, offer interpretable subgroup discovery, whereas causal machine learning methods, like causal forests, provide robust estimation in complex data structures. Bayesian approaches offer the added advantage of incorporating prior information and providing probabilistic inference.

Understanding the relative strengths and limitations of these methods is crucial for researchers and clinicians aiming to make **evidence-based, individualized treatment decisions**.

1.3 Objectives

This study aims to:

1. Compare recursive partitioning, causal machine learning, and Bayesian methods in terms of subgroup identification and treatment effect estimation.
2. Evaluate the accuracy, bias, variance, and interpretability of subgroup-specific treatment effect estimates.
3. Provide practical recommendations for applying these methods in clinical research settings.

1.4 Research Questions

- Which methods most effectively identify clinically meaningful patient subgroups?
- How do the methods differ in estimating subgroup-specific treatment effects regarding accuracy, bias, and uncertainty?
- What are the trade-offs between interpretability and predictive performance across these approaches?

2. LITERATURE REVIEW

2.1 Heterogeneous Treatment Effect Estimation

Heterogeneous treatment effects (HTE) reflect variability in how different patients respond to the same treatment. Accurately estimating HTE is critical for **personalized medicine**, enabling clinicians to tailor interventions based on patient-specific characteristics. Early approaches relied on subgroup analyses using stratified regression models, but these methods often suffer from **low statistical power**, **multiple testing issues**, and inability to detect complex interactions in high-dimensional data.

2.2 Recursive Partitioning Methods

Recursive partitioning techniques, particularly **model-based trees**, have been widely applied for HTE analysis. These methods iteratively split the dataset into subgroups based on covariate values to uncover treatment effect heterogeneity. Model-based trees are interpretable, allowing researchers to visualize subgroup characteristics and treatment responses clearly. Key studies demonstrate their utility in identifying clinically meaningful subgroups in both trial and observational data (Su et al., 2009; Seibold et al., 2016). However, their performance may be limited when handling large numbers of covariates or complex nonlinear interactions.

2.3 Causal Machine Learning Approaches

Recent advances in **causal machine learning** have enabled robust HTE estimation in high-dimensional settings. **Causal forests**, an extension of random forests for causal inference, estimate treatment effects by combining tree-based partitioning with propensity score adjustments, providing flexible modeling of interactions without overfitting (Athey et al., 2019). These methods excel in capturing subtle heterogeneity across complex covariates but may produce less interpretable subgroup structures compared to traditional trees.

2.4 Bayesian Approaches

Bayesian methods, such as **Bayesian Additive Regression Trees (BART)**, offer a probabilistic framework for HTE estimation. They can incorporate prior knowledge, quantify uncertainty in treatment effect estimates, and flexibly model nonlinear relationships and interactions (Hill, 2011; Chipman et al., 2010). Bayesian approaches have been successfully applied to both clinical trial and observational data, particularly when sample sizes are moderate, and covariate dimensions are high. The main limitation is computational intensity, especially with very large datasets.

2.5 Comparative Studies

Several studies have compared recursive partitioning and machine learning approaches for HTE estimation. While model-based trees are valued for interpretability, causal forests and Bayesian methods generally achieve higher predictive accuracy and better estimation of treatment effect uncertainty. However, evidence remains fragmented, with limited studies addressing **practical**

performance in high-dimensional clinical datasets, especially when multiple biomarkers and covariates are involved.

2.6 Gaps in the Literature

- Lack of systematic comparison of model-based trees, causal forests, and Bayesian models on the same high-dimensional datasets.
- Limited guidance on balancing **interpretability** with **estimation accuracy** for subgroup-specific treatment effects.
- Few studies assess computational efficiency and scalability of these methods in real-world clinical settings.

3. METHODOLOGY

3.1 Data Sources

- **Clinical Datasets:** The study will utilize high-dimensional clinical datasets, which may include randomized controlled trial (RCT) data or electronic health records (EHRs).
- **Patient Population:** Adult patients receiving a specific treatment or intervention, with inclusion criteria such as availability of key demographic, clinical, and biomarker data.
- **Covariates:** Demographic variables (age, sex, ethnicity), clinical variables (comorbidities, lab results), and biomarker measurements.
- **Outcome:** Primary clinical outcome of interest (e.g., treatment response, survival, or disease progression).

3.2 Methods for Subgroup Identification and Treatment Effect Estimation

1. Model-Based Trees (Recursive Partitioning)

- Uses **recursive partitioning** to split the dataset into subgroups based on covariates that maximize differences in treatment effects.
- Splitting criteria are based on statistical tests for parameter instability, ensuring subgroups reflect true heterogeneity.
- Advantages: Interpretable subgroup structures; intuitive visualization of patient splits.
- Limitations: May struggle with high-dimensional data and complex nonlinear interactions.

2. Causal Forests

- An ensemble learning method that builds multiple trees to estimate individualized treatment effects.
- Uses **propensity score adjustments** to correct for confounding in observational data.
- Provides **subgroup-specific treatment effect estimates** with uncertainty measures.
- Advantages: Flexible, robust to high-dimensional covariates, captures complex interactions.
- Limitations: Less interpretable than single trees; computationally intensive.

3. Bayesian Approaches (e.g., BART)

- Bayesian Additive Regression Trees model nonlinear relationships and interactions between covariates and treatment.

- Incorporates **prior information** and produces probabilistic treatment effect estimates.
- Advantages: Provides credible intervals and uncertainty quantification; flexible modeling.
- Limitations: Computationally demanding; requires careful tuning of priors and hyperparameters.

3.3 Evaluation Metrics

- **Accuracy of Subgroup Identification:** Measured by concordance with known or simulated subgroup structures.
- **Bias and Variance:** Assessment of the deviation of estimated treatment effects from true values.
- **Coverage Probability:** Proportion of times true subgroup treatment effects fall within estimated confidence or credible intervals.
- **Interpretability:** Qualitative assessment of how easily clinicians can understand subgroup definitions.
- **Computational Efficiency:** Time and resources required for each method.

3.4 Validation Strategy

- **Cross-Validation:** Repeated k-fold cross-validation to evaluate model performance and generalizability.
- **Simulation Studies:** Generate synthetic datasets with known treatment effect heterogeneity to benchmark methods.
- **Sensitivity Analysis:** Assess robustness to changes in sample size, covariate dimensionality, and noise levels.

3.5 Software and Implementation

- **R/Python Packages:**
- Model-based trees: `partykit`, `mob` (R)
- Causal forests: `grf` (R)
- Bayesian methods: `BART` (R) or `PyBART` (Python)
- **Computing Resources:** Use of high-performance computing or cloud-based environments to manage high-dimensional data processing.

4. RESULTS AND ANALYSIS

4.1 Subgroup Identification

- **Model-Based Trees:**
- Identified distinct patient subgroups based on key covariates such as age, biomarker levels, and comorbidities.
- Tree structures were interpretable, with clear decision rules showing which variables contributed to subgroup splits.
- Subgroups were clinically meaningful but relatively fewer in number due to conservative splitting criteria.
- **Causal Forests:**
- Detected more nuanced subgroups, capturing subtle interactions among covariates.
- Provided individualized treatment effect estimates across a continuum of patient characteristics.

- Subgroup boundaries were less interpretable compared to model-based trees but allowed finer granularity.
- **Bayesian Approaches (BART):**
- Generated probabilistic subgroup assignments rather than strict partitions, accommodating overlapping subgroup membership.
- Identified heterogeneity in treatment response with uncertainty estimates, which highlighted subgroups with high confidence of differential treatment effect.

4.2 Treatment Effect Estimation

- **Accuracy:**
- Causal forests and Bayesian methods showed lower bias and better recovery of true treatment effects in simulation studies compared to model-based trees.
- Model-based trees had slightly higher variance in treatment effect estimates, particularly for smaller subgroups.
- **Uncertainty Quantification:**
- Bayesian methods provided credible intervals for subgroup treatment effects, aiding clinical interpretation.
- Causal forests offered variance estimates from ensemble predictions.
- Model-based trees primarily relied on standard error approximations for treatment effect differences.

4.3 Comparative Performance

Method	Subgroup Interpretability	Bias	Variance	Coverage Probability	Computational Time
Model-Based Trees	High	Moderate	Moderate	Moderate	Low
Causal Forests	Moderate	Low	Low	High	Moderate
Bayesian (BART)	Moderate	Low	Low	High	High

Key Observations:

- Recursive partitioning is most interpretable but may underperform in complex high-dimensional settings.
- Causal forests balance predictive accuracy and subgroup granularity but at some cost to interpretability.
- Bayesian methods excel in uncertainty quantification and flexibility, though computational demands are higher.

4.4 Visualization and Interpretation

- **Tree Diagrams:** Illustrate model-based tree subgroup splits.
- **Forest Plots:** Compare subgroup-specific treatment effects across methods.
- **Heatmaps / Probability Plots:** Show Bayesian posterior probabilities for subgroup membership and treatment effect heterogeneity.

4.5 Summary of Findings

- Causal forests and Bayesian models are generally superior for high-dimensional datasets in estimating accurate subgroup-level treatment effects.
- Model-based trees provide interpretable decision rules useful for clinical translation.
- Choice of method depends on the trade-off between **interpretability**, **accuracy**, and **computational feasibility**.

5. CONCLUSION

5.1 Summary of Findings

This study conducted a **comparative evaluation** of recursive partitioning (model-based trees), causal machine learning (causal forests), and Bayesian approaches (BART) for **subgroup-specific treatment effect estimation** in high-dimensional clinical datasets. Key findings include:

- **Model-Based Trees:** Provided interpretable subgroup structures and clear decision rules but showed moderate bias and variance, particularly in complex datasets.
- **Causal Forests:** Offered robust estimation of individualized treatment effects with low bias and variance, capturing nuanced subgroup heterogeneity, though with reduced interpretability.
- **Bayesian Methods:** Excelled in uncertainty quantification and flexible modeling of nonlinear interactions, producing probabilistic treatment effect estimates, but required greater computational resources.

5.2 Practical Implications

- Recursive partitioning methods remain valuable for **clinically interpretable subgroup discovery**, particularly in settings where transparency is essential.
- Causal forests are recommended when **accurate estimation of treatment heterogeneity** is prioritized in high-dimensional data.
- Bayesian methods are advantageous when **uncertainty quantification and prior knowledge incorporation** are important for clinical decision-making.

5.3 Limitations

- Computational demands of Bayesian methods may limit their application in extremely large datasets.
- The study primarily focused on simulated and available clinical datasets; results may vary with different populations or outcome types.

- Interpretability trade-offs in causal forests and Bayesian approaches require careful consideration for clinical implementation.

5.4 Future Directions

- Explore hybrid approaches combining **interpretability of trees with predictive accuracy of causal machine learning**.
- Extend analysis to **multi-omics and real-world observational data** to improve personalized treatment strategies.
- Investigate methods for **dynamic treatment effect estimation**, considering longitudinal patient data.

This study contributes to the methodological toolkit for **precision medicine**, providing practical guidance on selecting suitable methods for estimating subgroup-specific treatment effects and enabling more informed, individualized treatment decisions.

6. REFERENCES

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